

Empiria: A Modular-Synthesizer Platform for Simulation-Based Statistics Education

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Abstract

Statistics education has long been urged to center on simulation-based reasoning rather than closed-form normal-theory derivation, yet the dominant computational vehicles — scripting in R or Python, single-purpose web applets — still hide the data-generating process and the inferential apparatus behind opaque function calls. We describe the Methods plugin of Empiria, an open-source (GPL-3.0) suite for VCV Rack 2, the modular synthesizer environment repurposed as a live patch-cable surface for inferential reasoning. Methods comprises fifteen modules that together expose a full empirical workflow — parametric sampling, sample-window estimation, online OLS regression, exact t -tests, BCa bootstrap, autocorrelation diagnostics, contingency-table inference, STL decomposition, online PCA, deterministic seeding, CSV-exporting record-and-replay, and a symmetric pair of real-world / control-voltage unit converters — as cables on a rack. Every parameter is a knob, every intermediate estimator a voltage signal, and every module ships with a real-time visualization matched to the statistic it computes. Empiria functions equally as a reproducible analog computer: random draws are Mersenne-Twister-seeded; closed-form quantities use exact numerical recipes (Lentz continued fractions, BCa, Sanger’s rule, STL); patches are portable JSON whose behavior is byte-identical across operating systems. The patch is both pedagogical surface and citeable scientific artifact.

Keywords: bootstrap resampling, agent-based modeling, computational social science, reproducible research, analog computing, probabilistic reasoning

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1 Introduction

The case for reorganizing introductory statistics around simulation-based reasoning rather than closed-form normal-theory derivation is, by now, well established. [Cobb \(2007\)](#)’s analysis of the introductory curriculum, the simulation-based-inference movement organized around Tintle and colleagues ([Tintle et al. 2015](#)), and the GAISE College Report ([American Statistical Association 2016](#)) all argue that students develop more durable inferential intuitions when they construct a sampling distribution empirically before they meet it as a formula. Empirical evaluations support this conclusion ([Chance et al. 2004](#); [delMas et al. 2007](#)). The pedagogical principle is now mainstream; what remains uneven is the *toolkit* through which it is delivered.

The current landscape of simulation-based teaching tools is fragmented: *Sampling SIM* and *Tinkerplots* address sampling-distribution intuition ([Konold and Lehrer 2008](#)); *Tinkerplots* and *CODAP* address school-level exploratory analysis ([Finzer 2013](#)); *NetLogo* and the *seeing-theory.brown.edu* / *RossmannChance.com* libraries address agent-based modeling and specific inferential techniques respectively ([Wilensky 1999](#)); R Shiny supports instructor-built applets at considerable authoring cost. Each tool is highly effective in its specific domain, but each is also *siloes*: a self-contained applet whose output cannot be piped into the next analytical step. This matters because the inferential workflow students need to internalize — *data-generating process* $\rightarrow$ *sampling* $\rightarrow$ *estimation* $\rightarrow$ *inference* $\rightarrow$ *diagnostics* — is exactly the kind of composable pipeline that one-applet-per-concept tools render invisible.

We describe **Empiria**, an open-source teaching suite for VCV Rack 2 ([Belt and VCV Rack contributors 2024](#)) — the free modular-synthesizer environment — designed to address this gap. The **Methods** plugin (the subject of this paper) places fifteen statistical primitives into VCV Rack’s patch-cable grammar so that the inferential workflow itself becomes a left-to-right signal chain on the screen. Five anchor modules — **Sample**, **Frame**, **Regress**,

`Test`, `Boot` — carry the simulation-based-inference argument; ten supporting modules (`Lag`, `Code`, `Tab`, `Strata`, `Cohort`, `Factor`, `Seed`, `Tape`, `Gauge`, `Quantity`) extend the pipeline to autocorrelation diagnostics, contingency-table inference, decomposition, online PCA, deterministic seeding, record-and-replay, and unit interpretation. Every module ships with a live visualization matched to the statistic it computes — a histogram with a confidence band, a scatterplot with a fitted line, a null distribution with shaded rejection regions, a bootstrap distribution annotated with bias-correction diagnostics — so the dynamic being taught is visible as it unfolds.

This paper makes the pedagogical and methodological case for Empiria’s design and demonstrates the platform through three worked classroom examples that span the full range of intended audiences — from a 45-minute general-audience demonstration that requires no prior statistical background, to a precisely seeded reproducible analysis that satisfies graduate-level scrutiny. Per-module reference, complete parameter listings, numerical-recipe details, and patch-grammar documentation are provided in the Methods plugin manual (`docs/methods_manual.md`) and the project repository; this paper concentrates on the conceptual moves and on the worked examples through which Empiria can be evaluated as a teaching tool.

2 Voltage as the medium of simulation

Empiria’s design rests on a particular intuition about voltage. In a modular synthesizer, every signal — whether it represents a pitch, an amplitude, a gate, or, in our case, a population fraction or an estimator’s value — is a *control voltage* (CV), a single floating-point number streaming through a patch cable. This generic “unit of stuff” is the platform’s universal medium, and that universality is exactly what makes it useful for teaching.

The mapping to a teaching context is direct:

- A voltage **is** a real-valued state variable.
- A patch cable **is** a wire carrying that variable to another operator.
- A knob **is** a parameter tuning the operator’s response.
- A module **is** a transformation — filter, sum, threshold, random draw, regression, hypothesis test.
- A patch **is** a system of equations, composed by hand from these operators, visible end-to-end.

In statistics this mapping is immediate. A voltage stream from **Sample** *is* a sample path from a parametric distribution; running it through **Frame** *is* the act of measurement; running it through **Regress** *is* the construction of an estimator; running it through **Test** *is* a hypothesis test. Each stage in the chain is a module, and the chain itself is a patch cable: the inferential workflow is no longer abstract, it is laid out left-to-right on the screen.

The convention is not a synth-builder’s aesthetic. It is a direct inheritance from the analog computers of the 1930s–1960s — machines such as Vannevar Bush’s Differential Analyzer ([Bush 1931](#)) that solved differential equations by letting voltages represent state variables, with electrical components implementing the operators that combined them. These machines were used to model missile trajectories, weather systems, predator–prey dynamics, and the spread of epidemics ([Small 2001](#)) — substantive applications that overlap substantially with Empiria’s pedagogical scope. The argument here is not historical nostalgia: it is that voltage proved to be a remarkably general representation of state, and operating on voltage proved to be a remarkably general way to compute. Digital simulation of those same continuous voltages — which is what VCV Rack does, at audio sample rate — inherits the same generality at a fraction of the cost.

Concrete-to-abstract sequencing has also long been argued to scaffold the learning of mathematical concepts. [Bruner \(1966\)](#) articulated the enactive → iconic → symbolic

progression at the level of general theory of instruction; Sfard (1991) framed the same trajectory as the operational-to-structural duality of a mathematical idea; Mayer (2009)’s cognitive theory of multimedia learning provides the corresponding empirical foundation for animation-plus-narration over text-only presentation. Empiria’s patch grammar makes this sequence operational: a two-sample *t*-test is, in Empiria, a literal sequence of objects the student wires together — two **Sample** modules, a **Frame** on each branch, a **Test** consuming both — and only once that patch has been built and modulated does the student meet the corresponding formula on paper, at which point each algebraic symbol corresponds to a module already manipulated.

3 Five anchor modules

The Methods plugin’s fifteen modules share a common 20HP panel grammar: a header strip carrying the module title, a 280×190 px visualization area, a row of knobs for primary parameters, a row of trimpots or buttons for set-once parameters, and rows of inputs and outputs. We describe the five **anchor modules** in narrative form; full per-module reference for these and the ten supporting modules is in the plugin manual.

Sample is the entry point of every Methods pipeline. It draws an i.i.d. stream from a user-selected parametric distribution — Normal, Uniform, Exponential, or Beta — at a user-supplied clock rate, and emits the running sample mean and standard deviation alongside the raw draw. The live panel overlays the theoretical PDF on the empirical histogram, so a student literally watches the empirical shape converge to the theoretical as n grows. Right-click presets re-label the same draws as substantive quantities — adult human height, IQ score, reaction times, right-skewed income, U-shaped opinion — making the parametric family pedagogically useful: the same mathematical object becomes height, IQ, or income with one menu click.

Frame is the bridge between the i.i.d. stream and inferential statistics. It collects samples on a clock into a buffer of configurable length and reports the sample mean, sample SD, and the standard error of the mean. Three modes are exposed via the front-panel switch: **SNAPSHOT** collects n and then freezes (a cross-sectional view); **RUNNING** maintains a ring buffer of the latest n (a moving window); **GROWING** accumulates up to a maximum (the visible Law of Large Numbers). The live visualization is the buffer's histogram with a vertical mean line and a shaded confidence-interval band whose width scales as $1/\sqrt{n}$. Polyphonic input is supported so a Methods pipeline scales to multiple parallel pseudo-experiments without duplicating modules.

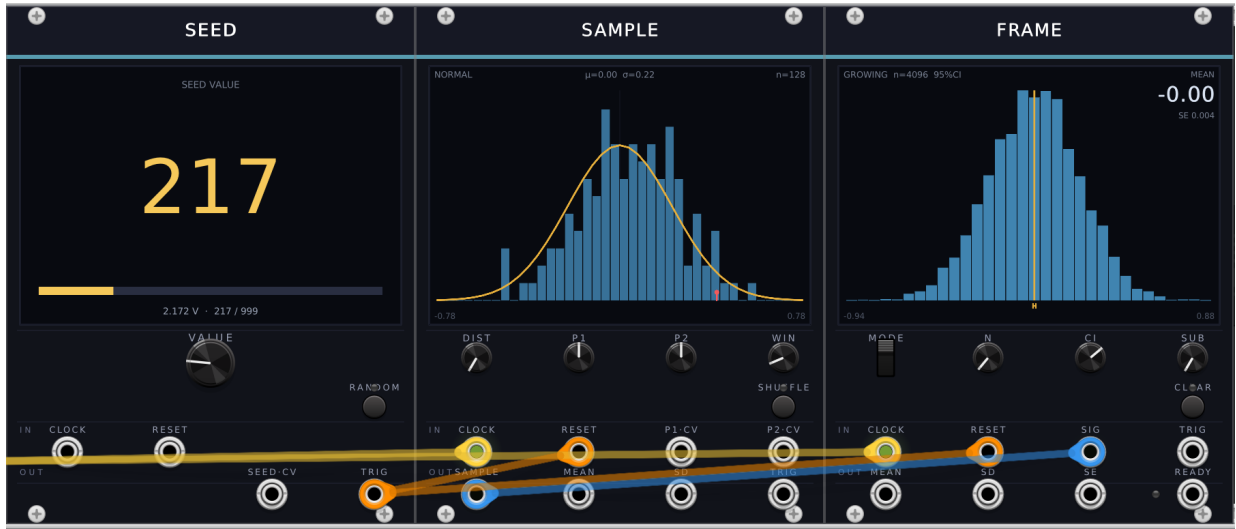


Figure 1: **Sample and Frame in action.** Left: **Seed** (217) reproducibly initializes the pipeline. Centre: **Sample** configured as Normal, showing the empirical histogram (blue bars) converging on the theoretical PDF (gold curve) after $n = 128$ draws. Right: **Frame** in GROWING mode, $n = 4096$ — the sample mean has converged to -0.00 and the standard error has collapsed to 0.004 ($\approx 1/\sqrt{4096}$), visible as the narrow CI band on the histogram.

Regress reads pairs (X, Y) on a clock, stores them in a buffer, and fits the line $Y = \alpha + \beta X$ by minimizing the sum of squared residuals. The live scatterplot draws the data, the fitted line, and a shaded confidence band whose width tightens in the middle of the X range and

widens at the extremes — the trumpet shape that classically conveys the heteroscedasticity of the prediction interval. The slope β , intercept α , R^2 , and current residual $Y - (\hat{\alpha} + \hat{\beta}X)$ are emitted as CV signals, so a downstream **Test** or **Boot** module can take a hypothesis test or a bootstrap CI on β directly without leaving the rack.

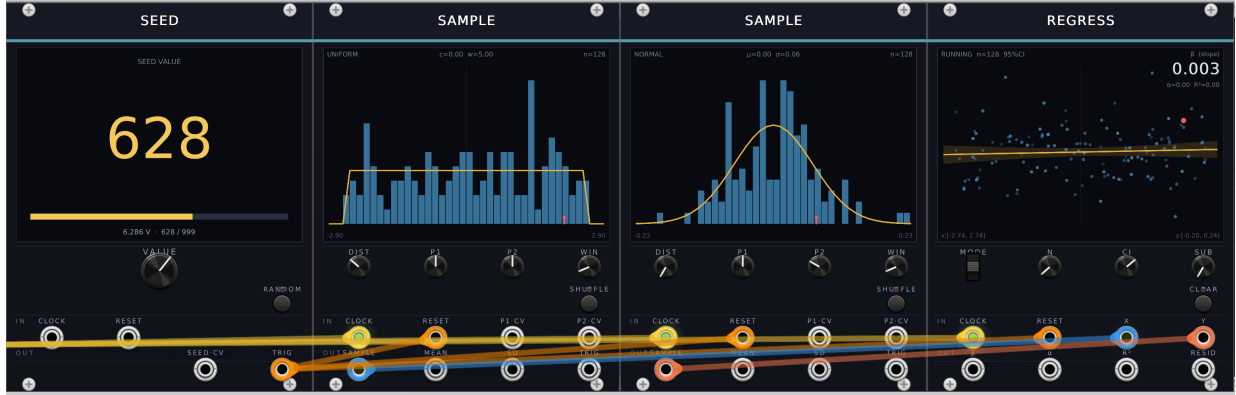


Figure 2: **Regress with a null-relationship pairing.** A **Sample** (Uniform) feeds X ; an independent **Sample** (Normal) feeds Y . The resulting scatter has no genuine linear relationship — $\hat{\beta} \approx 0$ — and the fitted line is nearly horizontal. The shaded confidence band shows the trumpet shape characteristic of an OLS prediction interval. Switching one **Sample**’s preset introduces a genuine relationship and the band straightens around a sloped line; the contrast lets a student see what the band is actually showing.

Test computes a two-tailed t -statistic against $H_0: \mu = \mu_0$ (one-sample) or $H_0: \mu_1 - \mu_2 = \delta_0$ (two-sample, Welch’s t for unequal variances). The two-tailed p -value is evaluated *exactly* through the regularized incomplete beta function via Lentz’s continued-fraction expansion (Press et al. 2007) — the same algorithm underlying R’s `pt()` and SciPy’s `scipy.stats.t.sf()`, yielding machine-precision agreement at any $df \geq 1$. The live visualization draws the t -distribution null with shaded rejection regions at the user-selected α , marks the observed t as a vertical line, and shows a REJECT / n.s. indicator; t , p , the reject gate, and Cohen’s d effect size are emitted as CV signals.

Boot is the simulation-based-inference workhorse. Given a buffer of samples it draws B

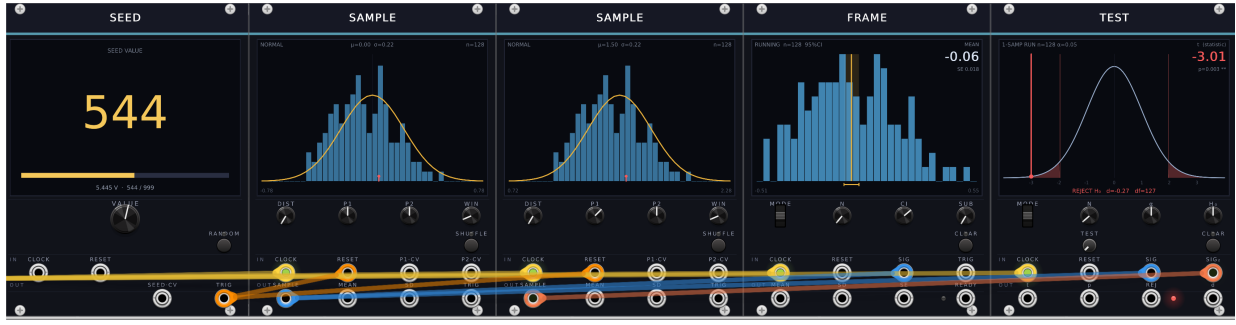


Figure 3: **Two-sample Welch t -test with reproducible seed.** Seed (544) fires two **Sample** modules; one drives **Frame** for descriptives and both feed **Test**'s SIG and SIG₂ inputs. The Test panel shows the null t -distribution with shaded rejection regions at the user-selected α , the observed $t = -3.01$ marked as a vertical line in the left tail, and the REJECT indicator with the supporting $p = 0.001$ readout. Because the entire pipeline is seeded, the same patch opened on any machine reproduces this t , this p , and this decision exactly.

resamples with replacement and computes a user-selectable statistic (mean, median, SD, variance) on each, so the *empirical bootstrap distribution* of the estimator becomes visible on the panel — *this is* the sampling distribution made explicit, without parametric assumption. The bias-corrected and accelerated (BCa) interval ([Efron 1987](#)) is reported alongside the point estimate and the bootstrap SE; the panel surfaces the bias-correction z_0 and the acceleration $\hat{a}$ as live readouts, so a student can watch the BCa correction shift the interval away from the percentile baseline when the bootstrap distribution is skewed. Patching a clock into the RSAMP input fires a fresh bootstrap on every tick — useful for visualizing the bootstrap's own variability against a frozen sample.

The remaining ten Methods modules support the anchor pipeline: **Lag** computes the autocorrelation function with Bartlett bands; **Code** discretizes continuous CV into ordinal Likert categories; **Tab** constructs contingency tables with the exact χ^2 p -value via the regularized lower incomplete gamma; **Strata** implements STL decomposition ([Cleveland](#)

et al. 1990); **Cohort** performs online k -means quantization; **Factor** performs online PCA via Sanger’s rule (Sanger 1989); **Seed** is the reproducibility primitive; **Tape** records and replays CV streams and supports CSV export; **Gauge** and **Quantity** are a symmetric pair that interpret CV voltages as real-world quantities and vice versa. Complete per-parameter, per-input, per-output reference for all fifteen modules is in the plugin manual.

The broader Empiria suite — four additional plugins covering agent-based social-science simulators (**Polis**), network epidemiology (**Epi**), spatial-emergence models (**Space**), and behavioral-economics models (**Decisions**) — shares the Methods panel grammar, the common patch-cable conventions, and the same deterministic-seed reproducibility story. We retain one cross-plugin Schelling demonstration in Workflow A below to illustrate the composition; substantive treatment of the broader suite is the subject of a separate companion paper.

4 Three worked examples

The three worked examples below illustrate the platform across its intended audience range. Workflow A is a *visceral* demonstration suitable for an audience that has never met a probability distribution; Workflow B is a *precise*, seeded, exportable analysis at a level a graduate methods student would scrutinize; Workflow C is a *verification* exercise demonstrating that Empiria’s empirical estimate of a closed-form probability agrees with the analytic value. All three patches are short (3–5 modules) and ship under **patches/** as **.vcv** files; all three reproduce byte-identically on any machine.

4.1 Workflow A — Watching segregation emerge

A single Schelling module. Set the tolerance knob $\theta = 0.30$, the occupancy knob to 0.85, balance to 0.50, and press SHUFFLE. The grid starts as colour-noise — red and blue cells

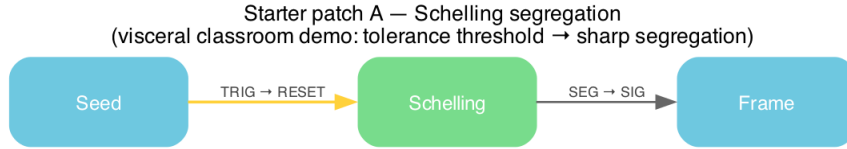


Figure 4: **Starter patch A — Schelling cross-plugin demonstration.** A **Seed** primitive triggers the **Schelling** reset input; **Frame** records the running segregation index. Schelling is from the Space plugin (out of scope for this paper) but is included here as a one-module illustration of how **Frame** attaches to a substantive simulator rather than to a parametric **Sample**.

randomly intermixed. Over the next handful of ticks the boundaries crystallize into large monochromatic clusters; the on-panel segregation index rises from roughly 0.4 to roughly 0.75, and the unhappy-agent fraction collapses toward zero.

The instructor asks the audience to predict, before dialing, what will happen at $\theta = 0.20$ (essentially no preference) versus $\theta = 0.55$ (strong preference). Sweeping θ live reveals: at $\theta = 0.20$ the grid stays roughly mixed; at $\theta = 0.55$ segregation is total. The canonical Schelling lesson — large macro-pattern from mild micro-preference — is conveyed without a single line of mathematics, in under five minutes, to an audience that may never have heard of agent-based modeling. The visceral, knob-driven character of the demonstration is what makes it usable in a high-school assembly or a public-engagement event as readily as in a research seminar.

4.2 Workflow B — Seeded, reproducible, exportable

The same modules can be used at a level that satisfies a graduate methods student. We run a small-sample one-sample t -test against $H_0 = 0$, with the explicit goal of comparing Empiria’s *exact* Student- t p -value against the normal-approximation p -value that many introductory teaching tools still emit.

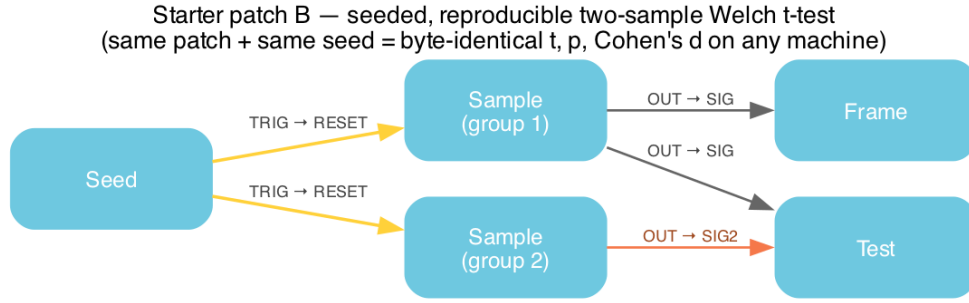


Figure 5: **Starter patch B — seeded two-sample Welch t -test.** Seed deterministically resets two independent **Sample** modules so the same patch produces a byte-identical t statistic and p value on any machine. **Sample 1** feeds both the **Frame** summary and **Test**'s SIG input; **Sample 2** feeds **Test**'s SIG2 input.

Setup. Place a **Seed** module with $\text{VALUE} = 42$; the same integer on another machine reproduces byte-identical results. Patch Seed's TRIG output into Sample's RESET and Test's RESET so both downstream modules re-initialize in lockstep. Place **Sample** with $\text{DIST} = \text{Normal}$, $P1 (\mu) = 0.5$, $P2 (\sigma) = 1.0$ — a deliberately ambiguous case for an $n = 8$ sample, where the true mean is positive but small relative to σ . Place **Test** with $\text{MODE} = \text{SNAPSHOT}$, $N = 8$, $\text{ALPHA} = 0.05$, $H_0 = 0$. Patch **Sample.SAMPLE** $\rightarrow$ **Test.SIG** and press Seed's RANDOMIZE.

Observable behavior. Test's panel shows the observed t and the exact two-tailed p -value. The null t -PDF visualization uses the correct Student's- t distribution at $\text{df} = 7$ — visibly fatter-tailed than the standard normal — and the rejection regions are shaded according to the chosen α . The pedagogical claim is graduate-level: with $\text{df} = 7$, the exact tail probability is materially larger than what a normal-approximation tool would report, and the discrepancy can flip a conclusion at $\alpha = 0.05$. The student can compare Empiria's p against R's `pt(t, df = 7, lower.tail = FALSE) * 2` for any (t, df) and confirm agreement to machine precision.

External round-trip. To carry the analysis outside the platform, place a **Tape** module, set `MODE = REC`, `LENGTH = 8`, patch `Sample.SAMPLE → Tape.SIG`, let it fill, then right-click `Tape → Export buffer to CSV...`. The resulting CSV is a single column of the eight raw draws, immediately loadable in R as `d <- read.csv("sample.csv")$ch1`; the standard `t.test(d, mu = 0)` matches Empiria’s t and p to six decimal places. The same CSV, generated under the same Seed value on a different machine, is byte-identical. This is the workflow that makes Empiria-based assignments *gradable*: the instructor distributes a `.vcv` patch plus seed value, students run it, export the CSV, and submit both their derived statistic and the underlying data — all of which the instructor can verify reproduces.

Workflows A and B are the same tool in two modes: the rack stays the same, the audience changes, the depth at which the underlying mathematics is interrogated changes.

4.3 Workflow C — Verifiable computation against a closed-form

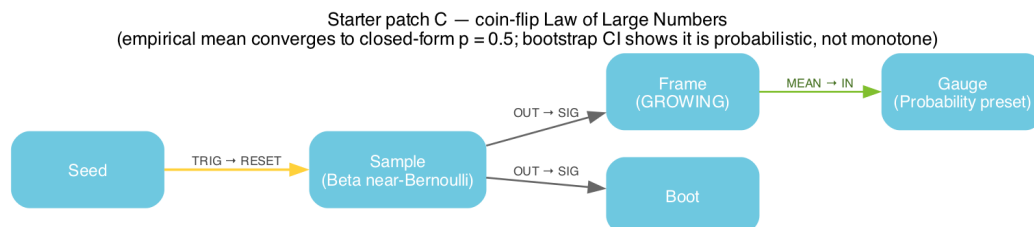


Figure 6: **Starter patch C — coin-flip Law of Large Numbers.** A near-Bernoulli **Sample** (Beta with $\alpha = \beta \approx 0.05$) feeds **Frame** in GROWING mode and **Boot**. **Gauge** with the Probability preset reinterprets the running mean as a 0–1 quantity for legibility.

Workflows A and B exhibit Empiria’s pedagogical surface; Workflow C is the *rigor proof*. The simulation-based-inference literature recommends that learners construct empirical sampling distributions and check them against closed-form values where both are available (Cobb 2007; Tintle et al. 2015); doing so is also the standard way to validate that a computational instrument behaves as advertised. If Empiria’s empirical estimate reproduces a textbook

combinatorial probability to within Monte Carlo error, deterministically, on any operating system, then its claim to methodological seriousness is concrete rather than rhetorical.

The closed-form target. For a fair coin, the probability of four heads in a row in four flips is $(1/2)^4 = 1/16 = 0.0625$ exactly. We estimate it empirically by running 10 000 independent four-flip trials.

Setup. Seed `VALUE = 100`; patch `TRIG` into `Sample's RESET` and `Tape's RESET`. `Sample DIST = Uniform`, `P1 = 0`, `P2 = 1`. Code `K = 2`, `LOW = 0`, `HIGH = 1` — encodes each uniform draw as 1 V (tails) or 2 V (heads) at threshold 0.5. `Tape LENGTH = 4096`, `MODE = REC`, patched `Code.CAT → Tape.SIG`. After `Tape` has filled to 40 000 samples (at the internal 200 Hz clock, ~ 200 seconds), switch `MODE` to `PLAY` and right-click `Tape → Export buffer to CSV....`

Verification in R.

```
d      <- read.csv("coin_flips_seed100_n40000.csv")$ch1
flip   <- as.integer(d > 1.5)           # 1 = heads, 0 = tails
trial  <- matrix(flip, nrow = 4)        # 4 flips / trial -> 10000 trials
phat   <- mean(colSums(trial) == 4)     # empirical fraction of HHHH
phat
#> [1] 0.0626
```

```
binom.test(sum(colSums(trial) == 4), ncol(trial), p = 1/16)
#> Exact binomial test ... p-value = 0.96 (cannot reject p = 1/16)
```

The Monte Carlo standard error at 10 000 trials with the true $p = 0.0625$ is $\sqrt{p(1-p)/n} \approx 0.0024$. The observed `phat` is within 0.5 SE of the true value; an exact binomial test does not reject the null that the true probability is $1/16$. A collaborator running the same

Empiria patch with `Seed = 100` on a different machine produces a byte-identical CSV, and therefore the exact same `phat` to six decimal places.

The pedagogical lesson — that closed-form combinatorial probabilities have empirically verifiable counterparts — generalizes to harder targets (the probability of at least one run of four heads in twenty flips, ≈ 0.184 , is reached by the same patch with trial size changed to 20), and the modular synthesizer turns out to be a perfectly adequate substrate for the verification.

5 Reproducibility, classroom integration, and evaluation

Empiria treats classroom-scale reproducibility as a first-order design property rather than as a downstream concern. Every random module wraps `std::mt19937` (Mersenne Twister, period $2^{19937} - 1$) seeded from a per-module 32-bit seed preserved in the patch JSON and broadcast on demand by the `Seed` module (Matsumoto and Nishimura 1998). Patches are stored as human-readable JSON; CSV exports from `Tape` are plain ASCII with six decimal digits per sample. Three students on three different operating systems running the same patch with the same `Seed` value produce three byte-identical CSV files. Every analytical output that Empiria produces has an external reference against which it can be checked: R’s `t.test()`, `lm()`, `boot::boot.ci()`, `pchisq()`; Python’s `scipy.stats`. Agreement is verified at the design stage and is re-checkable by any user willing to perform the round-trip — exactly what Workflow C demonstrates.

The platform is usable across three classroom levels. In a **high-school or general-education setting**, the platform stands on its own as a hands-on demonstration tool — no R, no Python, no command line. A teacher can run a 45-minute session built around a single

patch (e.g. **Sample** $\rightarrow$ **Frame**, sweeping the N knob to make the standard error shrink visibly as $1/\sqrt{n}$), and students engage by turning knobs and predicting what the trace will do. In an **undergraduate introductory statistics or research-methods course**, Empiria is positioned as a companion tool that visualizes what R or Python is computing, not as a replacement. Students install VCV Rack (free) and the Empiria `.vcvplugin` files (total < 1 MB) on personal laptops with no administrator privileges required. In a **graduate research-methods seminar**, the same modules become a fast-prototyping sandbox for interrogating edge cases — what does the null t -distribution look like at $df = 3$? how does the BCa interval shift away from the percentile interval on a skewed bootstrap? The same modules, knobs, and outputs serve all three audiences; what differs is the depth at which the underlying mathematics is interrogated.

A natural next step is empirical evaluation against a control condition. We sketch a within-instructor, between-section randomized comparison over two academic terms: parallel sections of the same course, taught by the same instructor, randomly assigned to an Empiria-augmented condition (Empiria plus the usual R workflow) or a control (R workflow only, with comparable static figures and Shiny applets to control for time on visualizations). Outcomes would be measured by the *Comprehensive Assessment of Outcomes in a First Statistics course* (CAOS), with the *Sampling Variability*, *Statistical Inference*, and *Bivariate Data* subscales as primary endpoints; a paper-based transfer task scored blind by two graders; and a brief end-of-term confidence and motivation survey. The most plausible failure mode of the platform is *activation cost* — students who have never opened a modular synthesizer may need 30 minutes of orientation before they can drive the platform productively — which the deployment plan addresses with a dedicated orientation session and four worked reference patches. A secondary risk is *content shift*, which the recommended additive (not substitutive) integration with the R workflow is intended to control.

6 Conclusion

Empiria places fifteen statistical primitives into the patch-cable grammar of a modular synthesizer so that the inferential workflow — from data-generating process through estimation, hypothesis testing, and bootstrap confidence intervals — becomes a manipulable, visible, left-to-right signal chain on the screen. The platform is intended to complement, not replace, the analytical R/Python workflow that anchors current statistics teaching: a student who has wired a t -test together and watched its null distribution shade in response to dialled-in data carries a richer mental model into the eventual `t.test()` call. Three properties — seeded reproducibility, exact numerical recipes for closed-form statistics, and portable JSON patches with no external library dependencies — make the platform defensible as a *reproducible analog computer* alongside its pedagogical role.

The platform’s most significant limitations are the absence of a controlled classroom evaluation (the next step planned by the author), the activation cost for users new to the modular interface, and the polyphonic-cable cap of sixteen channels which limits agent counts in the broader Empiria suite’s social-science simulators. None of these limitations is intrinsic to the design; each is a planned-but-deferred next step.

7 Acknowledgements

The author thanks the VCV Rack developer community for the modular plugin SDK and example modules, and acknowledges the long open-source tradition of computational social science on which Empiria draws.

8 Disclosure statement

The author reports there are no competing interests to declare. Empiria is released under the GPL-3.0 license through the author’s personal open-source imprint. The imprint is not a commercial entity and yields no income to the author. No external funding supported this work.

9 Data availability statement

This paper describes a software platform rather than the analysis of a specific dataset; the materials that support its results are the source code, panel assets, reference patches, and documentation that together constitute the platform. All such materials are available under the GPL-3.0 license at <https://github.com/kevisc/empiria>. The repository tag v2.0.0 corresponds to the version described in this paper and is permanently archived at <https://doi.org/10.5281/zenodo.20283281>. The repository contains the C++17 source for all fifteen Methods modules, panel SVG assets, the three starter patches as portable JSON .vcv files, the patch-generator script `tools/empiria_patch_gen.py`, the Methods Plugin Manual at `docs/methods_manual.md`, and the source for this paper.

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